



**Cambridge Instruments
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

EARTH TESTER · HAND-DRIVEN, METAL BODY

CIE/222M

Hand-driven analog earth tester - 3-terminal (earth resistance) and 4-terminal (earth resistance + soil resistivity). $\pm 5\%$ accuracy per IS: 9223/1979.

2kV Dielectric Test

1 MIN, CIRCUIT TO CONTAINING CASE

$\pm 5\%$ Accuracy

PER IS: 9223/1979

Battery-Free

HAND-CRANKED GENERATOR

Analog Earth Tester - Metal Body

A hand-driven generator-type earth tester for direct measurement of earth resistance in power circuits, telecommunications, railway electrification, and industrial installations - available in 3-terminal (earth resistance) and 4-terminal (earth resistance plus soil resistivity) types, with single up to five-range configurations.

- **3-terminal or 4-terminal** - earth resistance, or + soil resistivity
- **$\pm 5\%$ accuracy** - per IS: 9223/1979
- **Centrifugal governor** - constant voltage regardless of crank speed
- **Anti-magnetic aluminium body** - protected metallic flap over scale
- **Collapsible generator handle** - folds flush into casing
- **Single to five-range** - configurations available

Supplied as base instrument only - testing kit and carrying case available at extra cost.

Manufactured in India since 1963 - trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors.



CIE/222M · EARTH TESTER

01 Full Specification Sheet

PARAMETER	SPECIFICATION
Body	Anti-magnetic aluminium
Generator Type	Hand-driven generator, centrifugal governor
Terminal Types	3-terminal (earth resistance) or 4-terminal (+ soil resistivity)
Scale Length	80 mm (approx.)
Accuracy	$\pm 5\%$ (as per IS: 9223/1979)
Rotation Speed	160 r.p.m.
Range Configuration	Multiple range options available - specify at order
High Voltage Test	2 KV A.C. (r.m.s.), 1 min., circuit to containing case
Insulation Resistance Test	Better than 25 M Ω at 500V D.C.
Dimensions (L×W×H)	110 × 128 × 190 mm (approx.)
Weight	2.5 kg (approx.)
Warranty	12 months against manufacturing defects

Custom ranges and configurations can also be built against specific requirements.

02 Typical Applications

- Earth resistance measurement for power circuits and lightning protection systems
- Telecommunications and railway electrification earthing checks
- Domestic and industrial electrical installation earthing
- 4-terminal type: soil resistivity surveys for civil engineering projects (tunnels, drainage, sewerage)

03 Optional Accessories

- + Testing kit - 3-terminal: 2 spikes (45cm), 3 wire spools (10/20/30m), 1 hammer, bag
- + Testing kit - 4-terminal: 4 spikes (45cm), 4 wire spools (10/20/20/30m), 1 hammer, bag
- + Carrying case
- + Any other range - on request

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